

Machine Learning and its Applications

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Today's goals



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Module 1

Machine Learning Fundamentals I: Basic Concepts

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Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

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Module 3

Data Preparation

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- Module 9 Introduction
- ✔

Introduction to Module 9's Study Video • 2 min
- ✔

Why Data Preparation Is Not Optional in Machine Learning Reading • 5 min

- Module 9-1 Basic Data Cleaning Review
- ✔

Basic Data Cleaning Review Video • 38 sec
- ✔

Materials on Basic Data Cleaning Reading • 10 min
- ✔

Cleaning Data Is a Decision, Not a Checklist Reading • 5 min

- Module 9-2 Distributions, Outliers and Their Removal
- ✔

Distributions, Outliers and Their Removal: Gaussian Distribution, Skewness and Outliers Video • 6 min
- ✔

Distributions, Outliers and Their Removal: The Z-Score and IQR Method for Outliers Removal Video • 5 min
- ✔

Materials on Distributions, Outliers and Their Removal Reading • 10 min
- Outliers, Assumptions, and When “Cleaning” Becomes Damage Reading • 5 min

- Module 9-3 Data Transform
- ✔

Data Transform: Normalization, Standardization, Power Transform Video • 7 min
- ✔

Materials on Data Transform Reading • 10 min
- When Scaling Helps — and When It Quietly Breaks Your Model Reading • 5 min

- Module 9-4 Construct Training and Test Sets for Model Evaluation
- ✔

Construct Training and Test Sets for Model Evaluation: Methods, Implementation, and Stratified Sampling Video • 5 min
- ✔

Construct Training and Test Sets for Model Evaluation: Cross-Validation Video • 6 min
- ✔

Materials on Training/Test Sets Generation Reading • 10 min
- ✔

Evaluation Starts with the Split: When Performance Numbers Lie Reading • 5 min

- Module 9-5 Data Preparation Overview
- ✔

Data Preparation Overview Video • 3 min
- ✔

Materials on Data Preparation Reading • 10 min
- From Raw Data to Reliable Learning: What Actually Matters? Reading • 5 min
- Data Preparation and Evaluation Coach Dialogue • 5 min

- Module 9 Quiz
- ✔

Module 9 Quiz Practice Assignment • Grade: 100%

- Module 9 Assignments
- Module 9 Assignment 1 Graded App Item • 30 min
- Module 9 Assignment 2 Graded App Item • 30 min
- Module 9 Assignment 3 Graded App Item • 30 min
- ✔

XLS file for Module 9 Assignment Reading • 1 min
- Module 9 Recap Reading • 5 min

Module 4

Support Vector Machines

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- Module 10 Introduction
- Introduction to Module 10's Study Video • 1 min
- Why Support Vector Machines? Reading • 5 min

- Module 10-1 Support Vector Machine Fundamentals
- Support Vector Machine Fundamentals: Concepts

Module 9 Recap

 0:00 / 1:42 

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Over the past module, you have focused on **data preparation**, one of the most critical and easily underestimated stages in the machine learning workflow. You examined how real-world data is distributed, how outliers arise, and why they can seriously distort model training if left untreated. You learned practical methods for identifying and handling outliers using Z-scores and the IQR method, explored data transformation techniques such as normalization, standardization, and power transforms, and applied principled approaches for constructing training and test sets, including cross-validation and stratified sampling.

A key takeaway from this module is that **good machine learning performance does not start with sophisticated algorithms—it starts with well-prepared data**. Clean, appropriately transformed, and properly partitioned datasets reduce bias, improve generalization, and make model evaluation meaningful. At the same time, you should now recognize that data preparation alone does not guarantee a good model; it simply creates the conditions under which learning algorithms can work effectively.

With this foundation in place, we are now ready to move beyond preparation and into **modeling with more powerful algorithms**. In the next module, you will study **Support Vector Machines (SVMs)**—a widely used and theoretically grounded machine learning method that relies heavily on concepts you have just learned, including feature scaling, data distribution, and careful model evaluation. As you move forward, keep in mind that the quality of an SVM model, like any other, will depend not only on the algorithm itself but also on how well the data has been prepared.

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